Cpts 317. 9/12

Prob4. You need give me exaple languges L_1 & L_2 s.t.

$$L_1^* + (L_1^* L_2)^* \neq (L_1 + L_2)^*.$$

Take $L_1 = \{a\}$, $L_2 = \{b\}$.

$ba \notin$ LHS, but $ba \in$ RHS

Prob 3 (2). $(L_1^* L_2)^* L_1^* = (L_1 + L_2)^*$.

Proof. LHS $\subseteq$ RHS. ~~trivial~~ trivial.

RHS $\subseteq$ LHS. why?

$$(L_1 + L_2)^* = \bigcup_{\substack{k \geq 0 \\ i_1, \cdots, i_k \in \{1,2\}}} L_{i_1} \cdots L_{i_k}$$

It's concatenation of arbitrary L_1's & L_2's.

$$\subseteq \bigcup_{\substack{k \geq 0 \\ i_1 \cdots i_k \in \{1,2\}}} \text{LHS}$$

$$= \underbrace{L_1 \cdots L_1 L_2 L_1 \cdots L_2 \cdots L_2 L_1 \cdots L_1}_{k \text{ many}}$$

$$\subseteq L_1^* L_2 L_1^* L_2 \cdots L_2 L_1^* = \text{LHS}.$$

$$= \text{LHS}.$$

Summary:

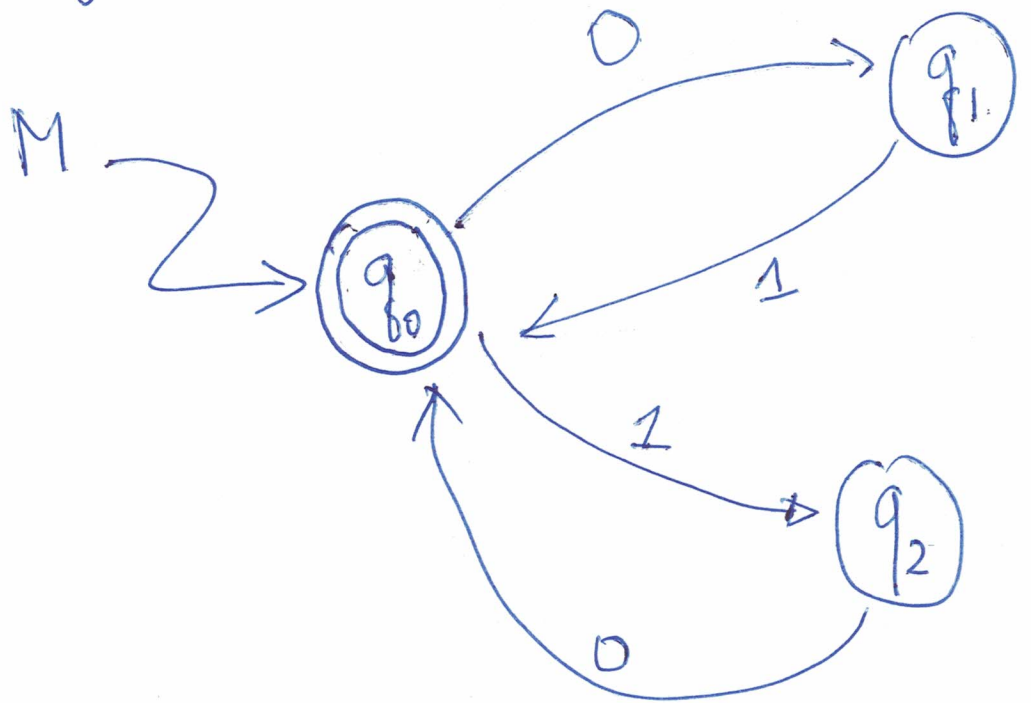
1. regExp / reglanguages. def's.

2. Write regExp. ⟸

3. structural Induction (proofs).

(Simplest class of languages,
‖
requirements).

Exaple FA:



An exaple run:

$$\rightarrow q_0 \xrightarrow{0} q_1 \xrightarrow{1} q_0 \xrightarrow{1} q_2 \xrightarrow{0} q_0 .$$

double-circled.

is called an accepting run.

0110 is accepted by M.

010110 is accepted by M

011 is not accepted by M.

001 is not accepted by M (since M crashes)

Challenge of today: how much memory does this exmple M have?

ANSWER:

$\log_2 3$ bits.

($\approx$ 2 bits).

1 bit of memory $\longrightarrow$ 2 states

2 bits of memory $\longrightarrow$ 4 states.

3 bits of memory $\longrightarrow$ 8 states.

4 bits of memory $\longrightarrow$ 16 states

$\vdots$

k bits of memory $\longrightarrow$ 2^k states.

$\log_2 n$ bits $\longleftarrow$ n states